

Towards a Vedic Algorithm for the Singular Value Decomposition: Part I - On Matrix Eigenvalues

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Abstract

In this note the authors outline a path towards a Vedic approach to the singular value decomposition which is important in quantum computing.

This work is dedicated to the progenitor of Vedic math.

Jagatguru Bharti Krishna Tirthaji introduced several techniques in Vedic mathematics. He mentioned that some volumes of his work were lost. In this note we attempt to extend his available techniques to finding the singular value decomposition of a matrix. We use a Vedic approach to construct the Singular Value Decomposition (SVD) by performing a breakdown of linear algebra concepts like eigenvalues, the Spectral Theorem, and the Polar Decomposition. This work will focus on the first concept, eigenvalues.

We begin by outlining how the linear transformation $(A - \lambda I)$ can be used to find the eigenvalues of A . Let's break this down step by step, leading up to finding the determinant and setting it to zero to solve for the eigenvalues using the Vedic method.

1 Defining the Transformation A and I

Given:

$$A(z_0, z_1, z_2, z_3) = (0, 3z_0, 2z_1, -3z_2)$$

and the identity transformation:

$$I(z_0, z_1, z_2, z_3) = (z_0, z_1, z_2, z_3).$$

2 Scalar Multiplication with the Identity Transformation

For a scalar c , the transformation $c \cdot I(z_0, z_1, z_2, z_3)$ is:

$$(c \cdot I)(z_0, z_1, z_2, z_3) = (c \cdot z_0, c \cdot z_1, c \cdot z_2, c \cdot z_3).$$

3 Defining $(A - \lambda I)$

The transformation $(A - \lambda I)(z_0, z_1, z_2, z_3)$ is given by:

$$(A - \lambda I)(z_0, z_1, z_2, z_3) = (0 - \lambda z_0, 3z_0 - \lambda z_1, 2z_1 - \lambda z_2, -3z_2 - \lambda z_3).$$

Simplifying this:

$$(A - \lambda I)(z_0, z_1, z_2, z_3) = (-\lambda z_0, 3z_0 - \lambda z_1, 2z_1 - \lambda z_2, -3z_2 - \lambda z_3).$$

4 Matrix Form of $(A - \lambda I)$

This transformation can be represented by the matrix $A - \lambda I$:

$$A - \lambda I = \begin{bmatrix} -\lambda & 0 & 0 & 0 \\ 3 & -\lambda & 0 & 0 \\ 0 & 2 & -\lambda & 0 \\ 0 & 0 & -3 & -\lambda \end{bmatrix}.$$

5 Finding the Determinant of $(A - \lambda I)$

The determinant of the matrix $A - \lambda I$ is given by:

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & 0 & 0 & 0 \\ 3 & -\lambda & 0 & 0 \\ 0 & 2 & -\lambda & 0 \\ 0 & 0 & -3 & -\lambda \end{vmatrix}.$$

To compute the determinant, we can use the standard algorithm which involves co-factors. The smallest co-factor, of size 2 by 2, can be evaluated using Vedic multiplication [1] and in a hierarchy upwards, each calculation can therefore be done in the Vedic way.

6 Setting the Determinant to Zero

More specifically, to find the eigenvalues, we set the determinant to zero to get an equation. Then we find the roots of the equation in the Vedic approach [2]. The eigenvalues of a matrix (or transformation) play a crucial role in the SVD. By finding the eigenvalues of the matrix $A^T A$, we can determine the singular values of the matrix A .

7 Conclusion

By leveraging the Vedic method in the derivation of eigenvalues, one can intuitively build the SVD of a matrix. The step-by-step nature of Vedic mathematics

aligns well with the sequential construction of SVD, making it possible to write out the entire procedure in a Vedic framework. The next steps are to construct the spectral theorem and polar decomposition in the Vedic framework, and this will be taken up in future work, which will bring us closer to the full SVD Vedic construction.

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References

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- [2] Swami Bharati Krishna Tirtha, Vasudeva Sharana Agrawala, et al. *Vedic mathematics*, volume 10. Motilal Banarsidass Publ., 1992.